Total No. of Questions : 8]

SEAT No. :

PE-4250

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S.E. (Electronics/E&Tc/Electronics & Computer/ Electronics (VLSI Design & Tech.)/E.C (A.C.T.))

SIGNALS AND SYSTEMS

(2019 Pattern) (Semester - IV) (204191)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the condidates:

1) *Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
2) *Assume suitable data if necessary.*
3) *Figures to the right indicate full marks.*

Q1) a) State the following properties of CT Fourier Series. **[6]**

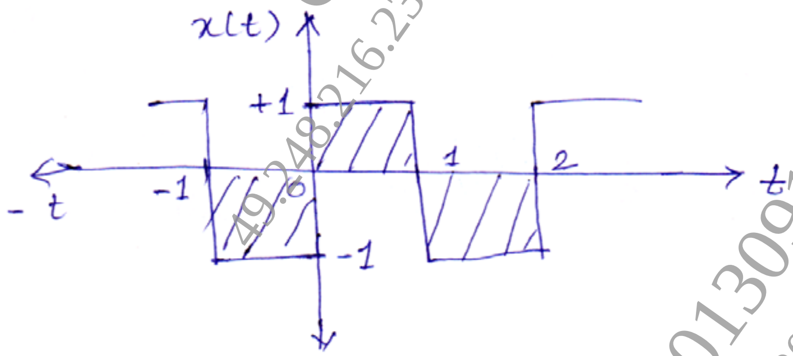
i) Linearity

ii) Time shifting

iii) Convolution

Also, give physical significance of above properties.

b) Find the Trignometric Fourier series for the bidirectional symmetric square waveform shown below. **[6]**



c) State and Explain Dirichlet conditions for the existance of Fourier Series. **[6]**

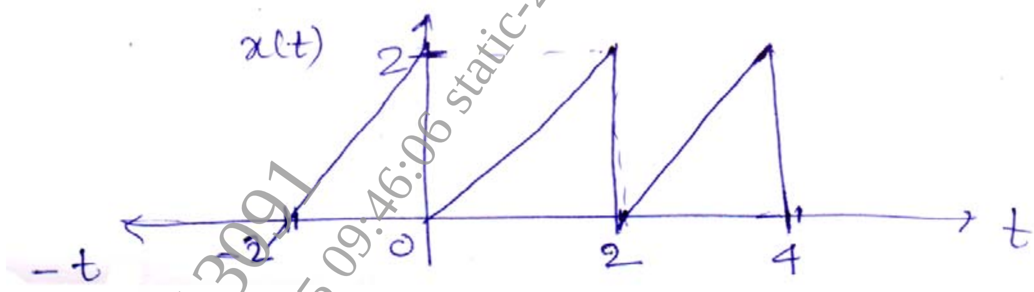
OR

Q2) a) Write a short note on Gibb's phenomenon for Fourier series. **[6]**

b) Check whether the two signals sin(wt) and cos(2wt) are orthogonal to each other. **[6]**

P.T.O.

c) Find the exponential FS for the periodic signal shown below. **[6]**



Q3) a) Find the Fourier Transform of the signal $x(t) = e^{-2t}u(t)$. Also sketch magnitude and phase response. **[6]**

b) Obtain the Fourier Transform using properties. **[6]**

i) $x(t) = e^{-2t}u(t + 2)$

ii) $x(t) = \delta(t) * \text{sgn}(t)$

c) State and explain the physical significance of following CTFT properties. **[5]**

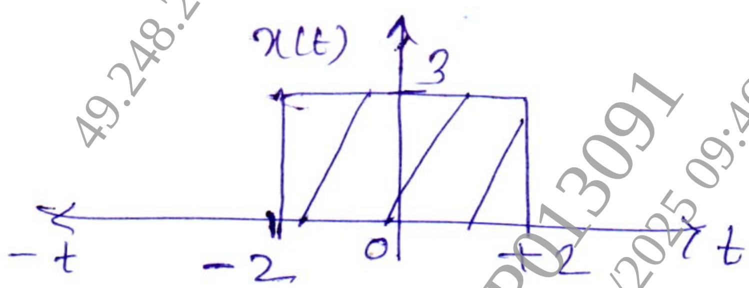
i) Time scaling

ii) Modulation

iii) Time Differentiation

OR

Q4) a) Find the Fourier Transform of a signal shown below. **[6]**



b) Find the Fourier Transform of $x(t) = \cos(w_0 t)$ using frequency shifting property of CTFT. **[6]**

c) Define magnitude response and phase response. Explain with one example. **[5]**

Q5) a) Define ROC in Laplace Transform. Also, state the properties of ROC. **[6]**

b) Find the Laplace Transform using properties of i) $x(t-2)$ and ii) $\frac{d}{dt}x(t)$ **[6]**

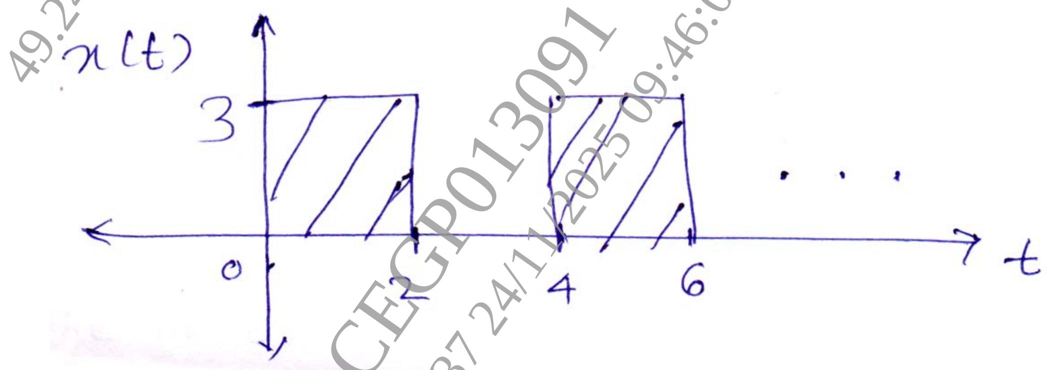
If $X(s)=\frac{2s}{s^2+2}$,

c) Find the Initial and Final values of $X(s)=\frac{5s+3}{s^2+2s+4}$ **[6]**

OR

Q6) a) Find the Inverse Laplace Transform of $X(s)=\frac{3}{(s+4)(s-2)}$ with given ROC, $-4 \le$ ROC $\le +2$ **[6]**

b) Find the Laplace Transform of given periodic signal. **[6]**



c) Compare Fourier Transform and Laplace Transform. **[6]**

Q7) a) State and Explain properties of PDF and CDF. **[6]**

b) Two dice are thrown simultaneously. Find the probability of following: **[6]**

i) Throwing a sum of 11

ii) Throwing two 7's

iii) Throwing a pair

c) Determine mean, mean square, variance for the given Probability Density Function (PDF). **[5]**

$F_x(x) = x^2$ for $0 \le x \le 1$

$= 0$ otherwise

OR

Q8) a) A certain random variable has the CDF. **[6]**

$$F_x(x) = \begin{cases} 0, & x \le 0 \\ kx^2, & 0 < x \le 10 \\ 100k, & \text{for } x > 10 \end{cases}$$

Evaluate k, $P(X \le 5)$ and $P(5 < X \le 7)$

b) A coin is tossed three times. A random variable X which represents the no. of tails. Obtained after the toss. Find the probabilities of X and plot the CDF. **[6]**

c) Write a short note on uniform Distribution Function. **[5]**